

Topic 3

Metals

Chemistry in Context

Liquidmetal Liquidmetal is the commercial name for a series of metal alloys. It has the properties of glass and metal and exhibits a number of distinct properties, including high tensile strength, excellent corrosion resistance, very little energy loss during a collision and excellent anti-wearing characteristics. Liquidmetal was introduced for commercial applications in 2003. It is used for golf clubs, watches and covers of cell phones. Researchers believe that liquidmetal will replace titanium in medical instruments, cars, the military and aerospace industry.



Do you recall?

1 What type of bonding exists in metals?
Metallic bond

2 In a metal, atoms are packed tightly together in a regular pattern. What is the name of this structure?
Giant metallic structure

3 Explain why metals are malleable.
As a metal is struck, the layers of positive metallic ions slide to new positions and are still bound by the sea of electrons.

In this topic, you are going to study the general properties of metals and the reactivity series of metals.

Unit

10 Occurrence and extraction of metals

Pre-study
A pre-study worksheet on this unit is available for download from *Jing Kung Chemistry Website*.

Unit Key Concepts

- The properties and uses of metals
- Different metal extraction methods
- Importance of recycling
- Conservation of metals

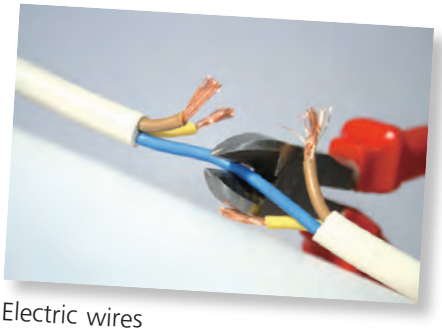
10.1 Metals in our lives

Metals play an important role in our everyday lives. Metals are included in many of the everyday articles that surround us. Often metals are N1 chosen for their durability. In certain applications metals are used because they are good conductors of heat and electricity. Fig. 10.1 shows some articles which are made of different metals.

N1 Refer to page T23.



Soft drink cans


Electric wires


Bridge


Jewellery N2


Light bulb filament


Artificial hip joint

Fig. 10.1 Some common articles made of metals

N2 Gold ornaments are usually made from gold alloys instead of pure gold because the alloys are stronger and harder than pure gold. The proportion of gold in its alloy is given by a carat number. 24-carat gold is pure gold. 18-carat gold contains 18 parts by mass of gold in 24 parts.

Discussion

Worksheet
A worksheet on this activity is available for download from *Jing Kung Chemistry Website*.

1 Identify the metal used for making each of the articles shown in Fig. 10.1.

2 For each article in the table below, list the properties that make the suggested metal suitable for making the article.

Article	Metal used	Properties that make the metal suitable for making the article
Soft drink can	aluminium	strong, light, malleable, resists corrosion
Electric wire	copper	very good conductor of electricity, ductile, resists corrosion
Bridge	iron	strong, easily shaped, cheap
Jewellery	gold	attractive yellow colour, resists corrosion, malleable and ductile
Light bulb filament	tungsten	very high melting point, gets 'white hot' without melting
Artificial hip joint	titanium	light but very strong, resists corrosion, easily shaped, biocompatible (non-poisonous and not rejected by the body)

Tensile strength measures the ability of a material to withstand a stretching force.

10.2 Uses of metals and their properties

Table 10.1 lists the properties of some common metals.

Table 10.1

Properties of some common metals

Metal	Melting point (°C)	Density (g cm^{-3})	Tensile strength	Electrical conductivity	Heat conductivity	Corrosion resistance
Aluminium	660	2.7	*	****	***	****
Copper	1 083	8.9	***	*****	*****	****
Gold	1 063	19.3	**	*****	****	*****
Iron	1 538	7.9	****	**	*	*
Silver	961	10.5	**	*****	*****	****
Titanium	1 660	4.5	*****	*	*	*****
Tungsten	3 410	19.3	**	**	**	****

(***** = best, * = worst)

tensile strength 抗張強度　　corrosion resistance 抗腐蝕性

Reference Website

Refer to the video clip 'Iron and steel':
http://www.rsc.org/learn-chemistry/resource/res00000023/iron-and-steel
(accessed July 2014)

A strong material can withstand a large force before it breaks.

Table 10.2 lists the uses of some common metals (Fig. 10.2). We can see that the uses of metals are closely related to their properties. N3

N3 When choosing an alloy for making fuse in electric plugs, the most important property is low melting point.

Table 10.2

Uses of some common metals and the reasons for their uses

Metal	Use	Reasons for use
Aluminium	making aeroplanes	light but strong, resists corrosion
	making window frames	strong, resists corrosion
Copper	making electric wires	very good conductor of electricity, very ductile, resists corrosion
	making cooking utensils	very good conductor of heat, very malleable, strong, resists corrosion, non-poisonous
Gold	making jewellery and coins	attractive yellow colour, resists corrosion (never loses its lustre), extremely malleable and ductile
	making electronic components	very good conductor of electricity, resists corrosion, extremely malleable, ductile
Iron	making bridges, ships and gratings	strong, malleable, ductile, cheap
Silver	making jewellery and coins	attractive shiny colour, resists corrosion, malleable, ductile
	making electric and electronic components	best conductor of electricity, resists corrosion, malleable, ductile
Titanium N4	making supersonic aircraft and space vehicles	light but very strong, very high melting point, resists corrosion
	making tooth implants and replacement hip joints	light but very strong, resists corrosion, can be easily shaped, *biocompatible* (non-poisonous and not rejected by the body)
Tungsten	making light bulb filaments	very high melting point, gets 'white hot' without melting

CE 2008 Paper 1 Q12(b)

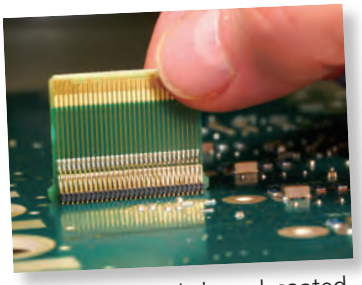
N4 Examination questions may ask about the uses of titanium. Titanium is also found in many consumer products, such as jewellery, watch cases, glasses, bicycles and clocks.



Aluminium is used to make overhead power cables



Copper is used to make cooking utensils



Electronic circuit board coated with gold



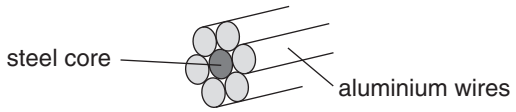
Titanium is used to make space vehicles

Fig. 10.2 Typical uses of metals

biocompatible 生物相容的

Practice 10.1

1 Aluminium is used for overhead power cables. These cables have a steel core.



a) The table below compares some properties of copper and aluminium.

Property	Copper	Aluminium
Density (g cm^{-3})	8.9	2.7
Cost (HKD per tonne)	57 200	14 600
Relative electrical conductivity	1.6	1.0

Use the information to explain why aluminium rather than copper is used for overhead power cables. Aluminium is cheaper. It is less dense so that the cables do not weigh so much. Its electrical conductivity is almost as good as copper.

b) Why do the cables have a steel core?
To increase the strength of the cables.

2 Titanium is used in *desalination* plants where fresh water is obtained from sea water. What properties of titanium make it suitable for this use?
Titanium resists corrosion in sea water.

10.3 Metals in the Earth's crust

The Earth's crust consists of many different substances. We get most metals from the Earth's crust. Fig. 10.3 shows the percentage by mass of elements in the Earth's crust. We can see that oxygen is the most abundant element in the Earth's crust. N5

N5 Refer to page T25.

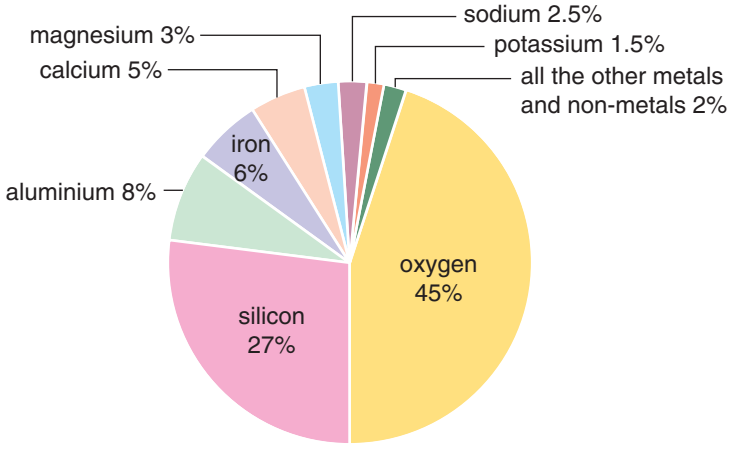


Fig. 10.3 Percentage by mass of elements in the Earth's crust

About 25% of the Earth's crust consists of metals. Aluminium and iron are the two most abundant metals.

desalination 海水淡化



Fig. 10.4 Gold exists as a free element in the Earth's crust

10.4 Extracting metals from their ores

Among the metals discovered so far, only a few metals such as gold and silver exist as free elements in the Earth's crust (Fig. 10.4). These metals can be extracted by physical methods. Table 10.3 lists some metals and their common ores. Most metals exist as compounds in their ores. We have to extract the metals from their ores using chemical reactions.

Table 10.3

Metals and their common ores

Metal	Common ore of the metal	Main metallic compound in the ore
Aluminium	*bauxite*	hydrated aluminium oxide ($Al_2O_3 \cdot 2H_2O$)
Copper	*copper pyrite*	copper(II) iron(II) sulphide ($CuFeS_2$)
Gold	as free element	free element of gold
Iron	*haematite*	iron(III) oxide (Fe_2O_3)
Lead	*galena*	lead(II) sulphide (PbS)
Magnesium	*magnesite*	magnesium carbonate ($MgCO_3$)
Mercury	*cinnabar*	mercury(II) sulphide (HgS)
Silver	*argentite* or as free element	silver sulphide (Ag_2S) or free element of silver
Sodium	rock salt	sodium chloride (NaCl)
Zinc	*zinc blende*	zinc sulphide (ZnS)

Fig. 10.5 shows haematite, a common ore of iron.



Fig. 10.5 Haematite contains iron(III) oxide

bauxite 鋁土礦　copper pyrite 黃銅礦　haematite 赤鐵礦　galena 方鉛礦　magnesite 菱鎂礦　cinnabar 硃砂
argentite 輝銀礦　zinc blende 閃鋅礦

Heating the ore alone

Mercury is poisonous. It vaporizes easily when heated. Hence the experiment should be carried out inside a fume cupboard.

This is the cheapest way to extract metals. We can extract mercury from cinnabar (containing mercury(II) sulphide) (Fig. 10.6a) by heating it in air.

mercury(II) sulphide + oxygen $\xrightarrow{\text{heat in air}}$ mercury + sulphur dioxide



Fig. 10.6a Cinnabar contains mercury(II) sulphide

To extract copper from copper pyrite (Fig. 10.6b), heat the ore with sand and air to produce copper(I) sulphide. Then convert the copper(I) sulphide to copper with a blast of air.

copper(I) sulphide + oxygen $\xrightarrow[\text{heat in air}]{\text{controlled}}$ copper + sulphur dioxide



Fig. 10.6b Copper pyrite contains copper(II) iron(II) sulphide

Reduction with carbon

Many metal ores are oxides. We can use carbon to extract these metals from their oxides (Fig. 10.7).

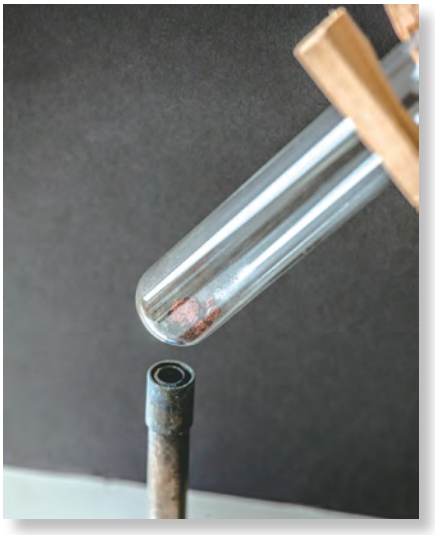


Fig. 10.7 Reddish brown copper formed in the reaction between carbon and copper(II) oxide

Video
Extracting copper from copper(II) oxide with carbon

We heat the metal oxide with carbon. The carbon removes the oxygen from the metal oxide to form carbon dioxide. The metal is also formed.

10.1
Extracting metals with carbon.

reduction

metal oxide + carbon ⟶ metal + carbon dioxide

The removal of oxygen from a metal oxide is called **reduction**. Carbon is a **reducing agent** that can remove oxygen from another compound.

At the temperature of a Bunsen flame (about 600 °C), carbon removes oxygen from lead(II) oxide.

CE 2008 Paper 2 Q34

lead(II) oxide + carbon $\xrightarrow{\text{heat}}$ lead + carbon dioxide N6

N6 Just mixing lead(II) oxide and carbon does NOT produce lead. Heating is required.

At 1 000 °C or above, carbon can also remove oxygen from zinc oxide and iron(III) oxide.

Carbon monoxide can also be used to reduce these metal oxides.

Converting sulphide ores into oxides

Before reduction, sulphide ores are usually converted to oxides by heating them in air, a process called **roasting**.

For example, zinc exists as zinc sulphide in its ore of zinc blende (Fig. 10.8). The zinc sulphide is converted into zinc oxide. Then zinc is extracted from its oxide by carbon.

zinc sulphide + oxygen $\xrightarrow{\text{roast in air}}$ zinc oxide + sulphur dioxide

zinc oxide + carbon $\xrightarrow{\text{heat}}$ zinc + carbon dioxide

Sulphur dioxide is an acidic gas. If it is allowed to escape, it will be converted into sulphuric acid by its reaction with oxygen and water in the atmosphere. The acid causes acid rain.



Fig. 10.8 Zinc blende contains zinc sulphide

reduction 還原作用　reducing agent 還原劑　roasting 煅燒

Extraction of iron — reduction of its ore by carbon monoxide in the blast furnace

A common ore of iron is haematite. It consists mainly of iron(III) oxide. In industry, iron is extracted from the ore in a special furnace called a **blast furnace**.

This is called a blast furnace because blasts of hot air are blown into the furnace.

N7 Refer to page T26.

During the extraction, a mixture of iron ore, coke (a form of carbon) and limestone is added from the top of the furnace (Fig. 10.9). At the same time, hot air is blown in at the bottom of the furnace. A chain of chemical reactions then occurs. Carbon monoxide is produced in the process.

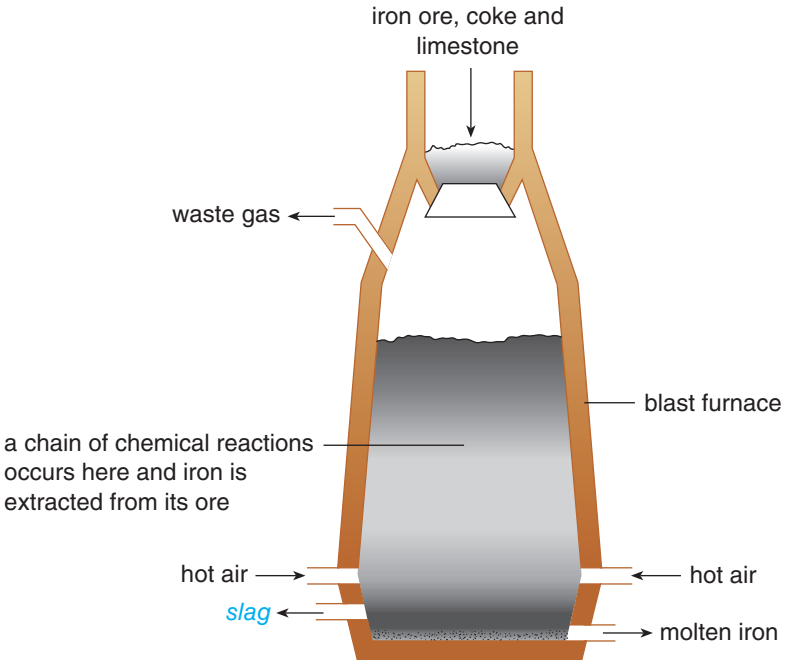


Fig. 10.9 Iron(III) oxide is reduced to iron in a blast furnace

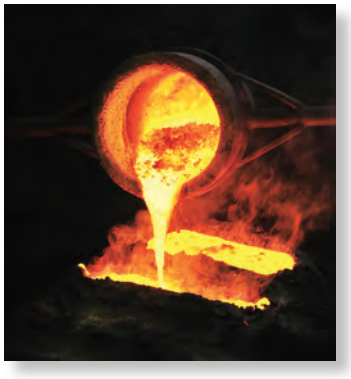


Fig. 10.10 Molten iron

The carbon monoxide reacts with iron(III) oxide in the ore, producing iron.

$$\text{iron(III) oxide} + \text{carbon monoxide} \xrightarrow{\text{heat}} \text{iron} + \text{carbon dioxide}$$

The high temperature inside the furnace melts the iron (Fig. 10.10). The molten iron runs to the bottom of the furnace and is tapped off. Only some molten iron is allowed to solidify in moulds. The rest is immediately used to make steel.

blast furnace 鼓風爐　　slag 爐渣

N8 Yellow street lamps contain sodium vapour.



Fig. 10.11 Bauxite contains aluminium oxide

Electrolysis

CE 2007 Paper 1 Q6(c)(i)
DSE 2012 Paper 1A Q9
DSE 2013 Paper 1A Q5
DSE 2014 Paper 1B Q4

To extract more reactive metals such as aluminium, magnesium and sodium, we have to use **electrolysis**, which is an expensive method. N8

The common ore of aluminium is bauxite (Fig. 10.11). It consists mainly of aluminium oxide. The electrolysis of bauxite is carried out in a large steel tank. The tank is lined with graphite, which acts as the negative electrode. Huge blocks of graphite hang in the middle of the tank and act as the positive electrodes (Figs. 10.12–10.13).



Fig. 10.12 Steel tanks where electrolysis of bauxite takes place

It would be too costly to melt aluminium oxide directly as its melting point is over 2 000 °C. Hence the oxide is first dissolved in molten cryolite (a less common aluminium ore). The melting point of the mixture is then lowered to just below 1 000 °C.

ASL 2007 Paper 2 Q3(a)(ii)
AL 2007 Paper 2 Q4(a)(ii)

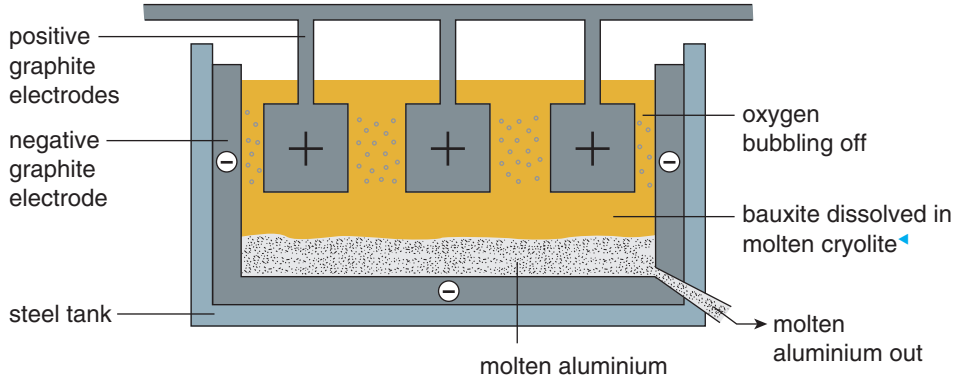


Fig. 10.13 Electrolysis of aluminium ore

During the process, aluminium forms at the negative electrode while oxygen gas bubbles off from the positive electrodes.

electrolysis 電解

CE 2009 Paper 1 Q2(b)(i)
DSE 2014 Paper 1B Q4

Example 10.1

Q Carbon can be used to extract metals from certain metal oxides.

a) i) Suggest how copper can be extracted from copper(II) oxide using carbon. State the expected observations.

ii) Write a word equation for the reaction involved.

b) Explain whether the method of carbon extraction can be used to distinguish between copper(II) oxide and silver oxide.

c) Explain whether carbon can also be used to extract magnesium from magnesium oxide.

CE 2009 Paper 1 Q2(b)(ii)

A a) i) Mix carbon powder with copper(II) oxide, and heat the mixture strongly.

A reddish brown powder is obtained.

ii) copper(II) oxide + carbon $\xrightarrow{\text{heat}}$ copper + carbon dioxide

b) Yes

Both copper and silver can be extracted from their oxides by heating with carbon powder.

Copper(II) oxide produces a reddish brown powder while silver oxide produces a grey powder.

c) No

Magnesium is a very reactive metal.

Do you know

Is it worth to extract the metal?

Ores contain metallic compounds in a concentrated form. However, it does not necessarily mean that it is worth the cost and effort to extract the metal. The value will depend on

- how concentrated the ore is;
- how easy it is to get the ore out of the ground;
- how easy it is to extract the metal from the ore;
- what price a company gets for the metal extracted;
- the long-term environmental clean-up costs.

The economics of the extraction process depend on the price of the metal and the demand for it.

10.5 Year of discovery of metals and the ease of extraction

In the Earth's crust, aluminium is more abundant than iron. However, iron was discovered much earlier than aluminium. Why?

Discussion

Worksheet
A worksheet on this activity is available for download from *Jing Kung Chemistry Website*.

The following table lists the year of discovery of some metals.

Metal	Year of discovery	Common ore	Common extraction method
Gold	8000 B.C.	as free element	mechanical separation
Mercury	8000 B.C.	cinnabar	heating in air
Copper	6000 B.C.	copper pyrite	controlled heating in air
Lead	3500 B.C.	galena	sulphide $\xrightarrow{\text{roast in air}}$ oxide; oxide $\xrightarrow{\text{heat with carbon}}$ lead
Iron	1200 B.C.	haematite	oxide $\xrightarrow{\text{heat with carbon}}$ iron
Sodium	1807	rock salt	electrolysis of molten ore
Aluminium	1826	bauxite	electrolysis of molten ore

1 Complete the last two columns of the table.

2 Is there any relationship between the year of discovery and extraction method of the metals?

The sequence of discovery of various metals relates closely with the ease of extracting the metal from their ores. Metals that are difficult to extract were discovered later.

Historically, the sequence of discovery of various metals relates closely with the ease of extracting the metals from their ores.

Metals such as aluminium and sodium are difficult to extract. These metals were discovered after the invention of electrolysis in 1800. Thus, aluminium was discovered later than iron although it is more abundant than iron in the Earth's crust.

Worksheet

Two related worksheets with answers are available for download from *Jing Kung Chemistry Website*. Worksheet ① consists of additional questions for promoting active reading. (Note: Questions in this worksheet also cover those in the textbook.) Worksheet ②, requiring students to indicate the relationship among various terms by drawing, may help the less able readers improve their reading skills.

Chemistry Magazine

Metals as part of human civilization

The development of human civilization has relied heavily on the discovery of metals. Historians divide particular periods of human history according to the materials mainly used, such as the Copper Age (5000–3000 B.C.), the Bronze Age (3000–1000 B.C.) and the Iron Age (1000 B.C.–A.D. 1).

The Copper Age

The first metal to be extracted from an ore was copper. Copper oxide was heated with charcoal (mainly carbon) and globules of copper resulted. Copper was commonly used for ornamental purposes. However, copper lacks the strength for making weapons.

The Bronze Age

Then, ancient metallurgists discovered that heating ores of copper and tin with charcoal could produce bronze which was significantly stronger than pure copper. Bronze was used to make tools and weapons in the Bronze Age (Fig. 10.14).



Fig. 10.14 A bronze bucket

The Iron Age

Tools and weapons made of iron were harder than those made of bronze. However, extraction of iron from its ore requires a higher temperature than that required for the extraction of copper. It was not until the invention of the charcoal furnace about 1200 B.C. that iron could be extracted from its ore effectively. Around 1000 B.C., iron was extensively used for making tools and weapons (Fig. 10.15).



Fig. 10.15 An iron sickle

Historians usually 'end' the Iron Age at about A.D. 1 and start talking about the Modern Age. However, iron is still the most widely used metal today.

Questions

1 Historically, gold and silver were used before copper and bronze, but mainly for ornamental purposes.

a) Why were they used before copper and bronze? Gold and silver exist as free elements and need no chemical extraction.

b) Why were they NOT used for tools and weapons? Gold and silver is too soft.

2 Suggest why iron is

a) the most extensively used metal; and

b) rarely used in its pure form.
Iron rusts easily.

3 In early times *pewter* was 75% tin and 25% lead. In more recent times it has been 85% tin, 7% copper, 6% bismuth and 2% antimony. The physical properties of both types of pewter are very similar. Suggest TWO reasons why there was a change in the composition of pewter.
To eliminate the risk of lead poisoning.
Bismuth and antimony were not known in ancient times.

2 a) Iron is the second most abundant metal in the Earth's crust.
Iron is strong and cheap.
It is the main constituent of steel. Steel is a very widely used building material.

pewter 錫鑞

10.6 How long will metal reserves last?

We have learnt that metals play an important role in our daily lives. However, metal deposits are non-renewable resources and they will run out soon if exploited at the present rate. Table 10.4 lists the year the reserves of some metals will last.

Table 10.4

Estimated reserve life of some metals

Metal	Reserve will last until
Copper	2045
Aluminium	2166
Tungsten	2050
Iron	2120
Silver	2030
Platinum	2027
Zinc	2037
Tin	2030

Worksheet **Assessment Form**

A worksheet and an assessment form on this activity are available for download from *Jing Kung Chemistry Website*.

Debate

'Aluminium should not be used to make food and beverage containers.'

You are going to hold a debate about the above motion. Your teacher will inform you in advance so that you may prepare for the debate. Two class periods should be assigned for the debate. Your teacher will inform you of further details in due course.

Refer to page T27.

10.7 The life cycle of metals

Mining, mineral processing and metal extraction produce many valuable metals for making products, but these activities can also have a serious impact on the environment. Fig. 10.16 shows the life cycle of a metal article.

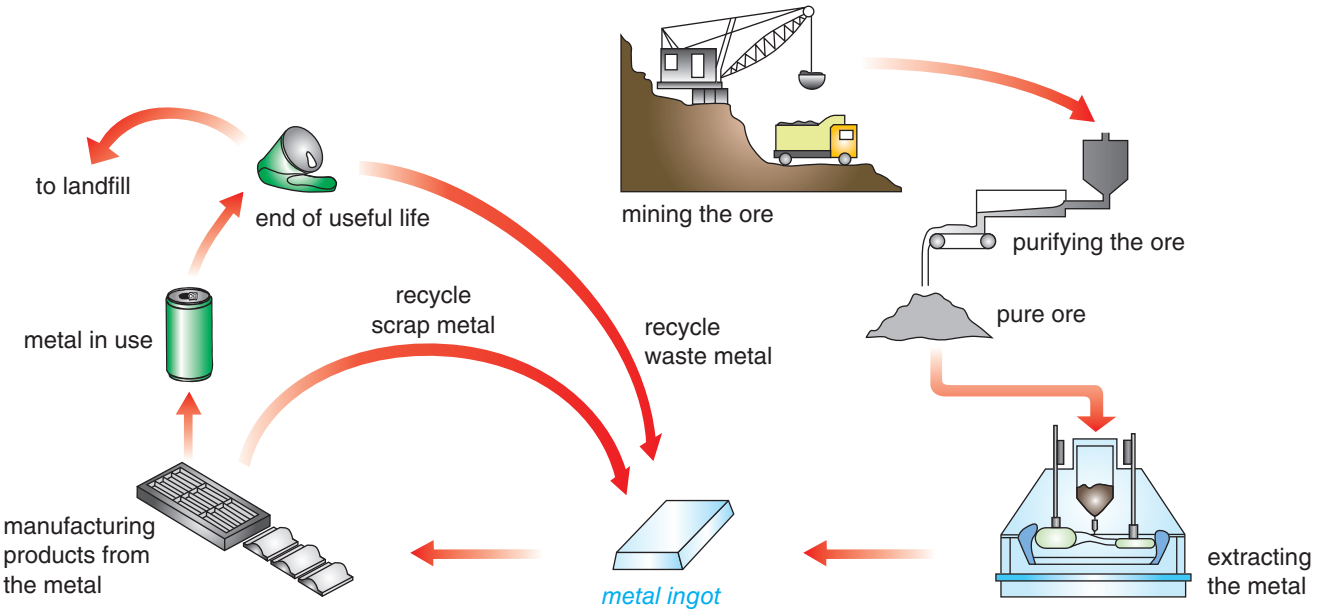


Fig. 10.16 The life cycle of a metal article



Fig. 10.17 Mining copper ores can leave large holes in the ground

Mining

There are basically two main types of mining: surface mining and underground mining.

Mining produces large volumes of waste rock and can leave very large holes in the ground (Fig. 10.17).

Purifying the ore

All metal ore processing produces waste. Waste tips may contain poisonous metals, such as copper and lead, that dissolve in rainwater and enter the soil.

Metal extraction

The metal extraction processes consume energy and water. Some air pollutants, such as sulphur dioxide, are given off during the processes.

metal ingot 金屬鑄塊

Metals in use

Careful choice of metals can reduce the environmental impact of mining. For example, lighter cars mean less fuel consumption and emission of harmful gases, as well as less wear and tear on roads. We can design lighter cars by replacing steel with lighter metals such as aluminium or with plastics and other new materials.

Recycling

Recycling of metals means melting down used and *scrap* metals, and using them again.

Scrap metals are generated by metalworking industries due to processes such as cutting and *boring*, etc.

The two most widely used metals are iron and aluminium. Both metals are now extensively recycled.

10.8 Recycling of metals

Fig. 10.18 shows various stages in the recycling of aluminium cans. At the recycling centre, aluminium cans are cut into chips, then the chips are melted and poured into *ingot moulds*.

The interior *lacquer* coating and the outside product display printing inks are first removed from cans.

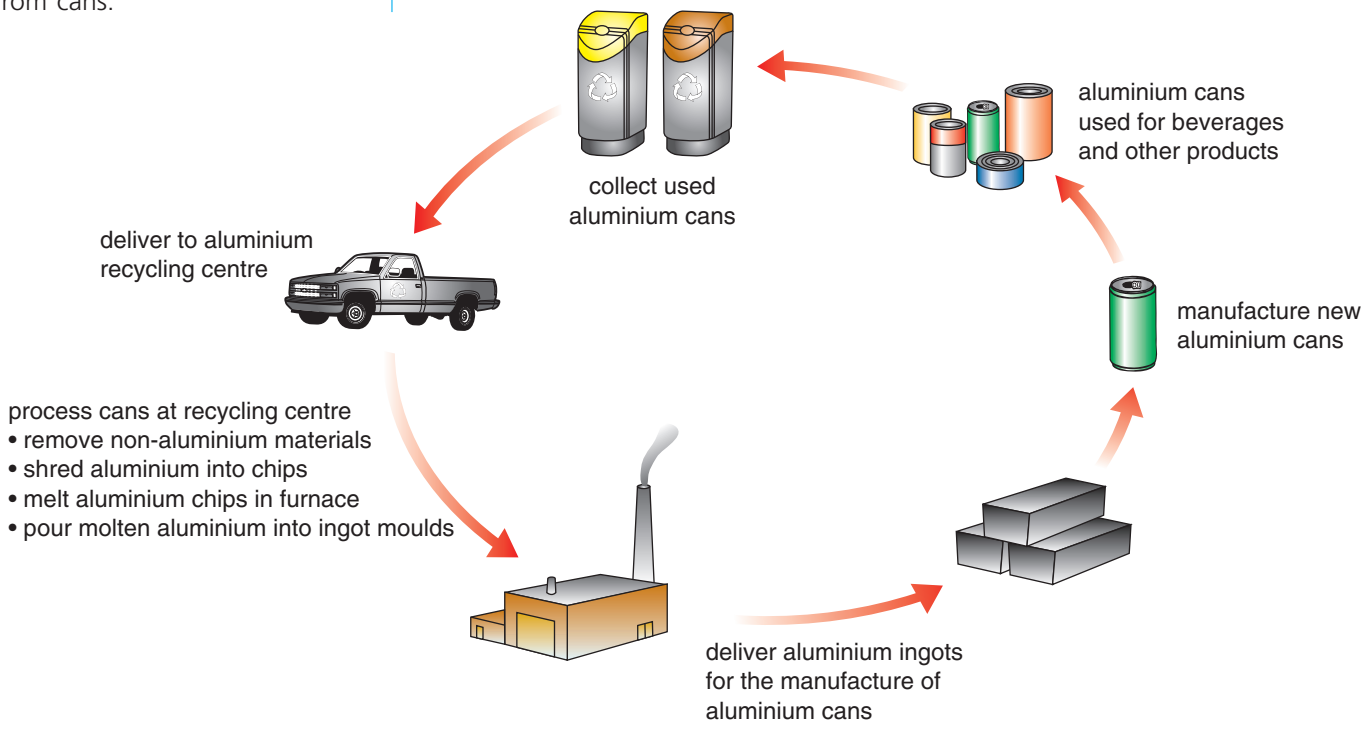


Fig. 10.18 Stages involved in the recycling of aluminium cans

recycling 循環再造　scrap 碎料　boring 鑽屑　lacquer 光漆　ingot mould 錠模

Video
The pros and cons of recycling

Importance of recycling

Recycling is very important because:

1 **Metal resources are limited** It takes four tonnes of bauxite ore to make one tonne of aluminium.

2 **Fuel and other resources can be saved** Less energy is used in recycling metals than in extracting the metals from ores (Table 10.5).

ASL 2007 Paper 2 Q3(a)(iii)
AL 2007 Paper 2 Q4(a)(iii)

Table 10.5

Energy required for the production of 1 kg of some metals

Metal	Energy required (in 10^6 J)	
	from ores	from recycled material
Aluminium	200	7
Copper	70	4
Mild steel	40	8

Other resources can also be saved by recycling. For example, 200 000 litres of water are needed to make one tonne of steel from iron ore. Only 120 000 litres of water are needed if scrap steel is reused.

3 **Environmental impact can be reduced** Recycling reduces the amount of metal waste that has to be disposed of. This is a major advantage for big cities where landfill sites are becoming hard to find.

Thinking in terms of greenhouse gases, melting scrap iron does not produce carbon dioxide whereas extracting iron from its ore does.

4 **Public awareness of conservation can be raised** The government encourages waste reduction and recycling activities (Fig. 10.19). People are becoming more aware of the metal conservation issue.



Fig. 10.19 A government poster which encourages source separation of waste for recycling

Problems in recycling

The main problem in recycling is cost. The main costs of recycling include:

1 the cost of collecting the waste metal;

2 the cost of transporting waste metal to the recycling centre;

3 the cost of melting and purifying the metal; and

4 the cost of disposing waste from the recycling process.

People recycle gold because gold has a high value. However, it may not be economical to recycle a cheap metal.

Other problems in recycling include the difficulty of getting people to save metal waste and the difficulty of sorting the metal waste. Some alloys cannot be sorted by inspection.

Worksheet
A worksheet on this activity is available for download from *Jing Kung Chemistry Website*.

Discussion

An international company makes the following claim: 'Aluminium products play a significant role in everything we do today. So, although the production of aluminium consumes resources and energy, the metal makes a major contribution to our quality of life.'

Draw a table to list the benefits and costs of our use of the metal. Do you agree with the claim made by this company? Refer to page T28.

10.9 Conservation of metal resources

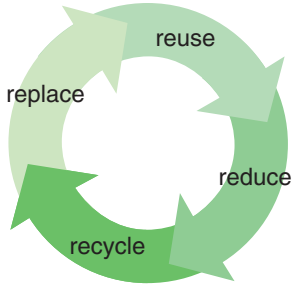


Fig. 10.20 Methods for conserving metal resources

Metal resources are limited. We must look for ways to conserve these valuable resources. The methods are **Reuse**, **Reduce**, **Recycle** and **Replace** (Fig. 10.20).

1 **Reuse** Many metal articles can be reused. For example, cookie containers can be reused for storing food. Used barbecue forks can be cleaned for reuse. Broken equipment can be repaired for reuse.

2 **Reduce** For example, use less packaging for mooncakes can help conserve metal resources.

3 **Recycle** Iron and aluminium are now extensively recycled (Fig. 10.21).



Fig. 10.21 Recycling bins for used aluminium cans

4 **Replace** Scientists have already used new materials to replace metals. For example, we have replaced copper by *optical fibres* for making telephone wires.

reuse 再用 reduce 減少 replace 取代 optical fibre 光纖

However, for some uses metals cannot be replaced by other materials. For example, we always use metals when electrical conductivity is required. We must also remember that eventually even the substitutes themselves may run out.

Discussion

Worksheet
A worksheet on this activity is available for download from *Jing Kung Chemistry Website*.

1 What are the advantages and disadvantages of replacing

a) a copper cooking pot with a glass one?

b) an aluminium soft drink can with a plastic one?

c) coins with money notes?

2 Hundreds of thousands of boxes of mooncakes are consumed by Hong Kong people every year. As a result, a considerable quantity of waste metal containers and packaging materials is generated. The Environmental Protection Department has issued guidelines to encourage mooncake manufacturers to use more environmentally friendly packaging design for mooncakes.

Apart from the efforts made by the mooncake manufacturers, suggest other measures you can take to help reduce waste arisen from mooncakes.

1

	Advantage(s)	Disadvantage(s)
a) Replacing a copper cooking pot with a glass one	• better appearance (transparent) • helps to conserve copper resources • no corrosion occurs	• easily broken
b) Replacing an aluminium soft drink can with a plastic one	• lighter • helps to conserve aluminium resources	• easily scratched
c) Replacing coins with money notes	• lighter • helps to conserve metal resources • no corrosion occurs • less storage space required	• not as durable as coins • more easily forged • easily damaged

The teacher may accept other sensible answers.

2 Measures students may suggest:
- Do not buy mooncakes with excess packaging.
- Recycle waste metal containers and other packaging materials.

Have you mastered?

Key terms

aluminium 鋁 3 titanium 鈦 3 tungsten 鎢 3 tensile strength 抗張強度 3

corrosion resistance 抗腐蝕性 3 bauxite 鋁土礦 6 copper pyrite 黃銅礦 6

haematite 赤鐵礦 6 galena 方鉛礦 6 magnesite 菱鎂礦 6 cinnabar 硃砂 6

argentite 輝銀礦 6 zinc blende 閃鋅礦 6 reduction 還原作用 8 reducing agent 還原劑 8

roasting 煅燒 8 blast furnace 鼓風爐 9 electrolysis 電解 10 recycling 循環再造 16

reuse 再用 19 reduce 減少 19 replace 取代 19

Checklist

After studying this unit, you should be able to

- ☐ relate the uses of metals to their properties;
- ☐ state the sources of metals and their occurrence in nature;
- ☐ explain why extraction of metal is needed;
- ☐ understand that the extraction of metals involves reduction of their ores;
- ☐ describe and explain the major methods of extraction of metals from their ores;
- ☐ relate the ease of obtaining metals from their ores to the reactivity of the metals;
- ☐ deduce the order of discovery of some metals from their relative ease of extraction;
- ☐ write word equations for the extraction of metals;
- ☐ describe metal ores as a finite resource and hence understand the need to recycle metals;
- ☐ evaluate the recycling of metals from social, economic and environmental perspectives.

(Put a '✔' in the box if you have acquired the knowledge concerned.)

Summary

1 Aluminium and iron are the two most abundant metals in the Earth's crust.

2 The following table summarizes different methods of metal extraction.

Extraction method	Example	Description and word equation(s) for the example
Heating the ore alone	extracting mercury from cinnabar (containing mercury(II) sulphide)	mercury(II) sulphide + oxygen $\xrightarrow{\text{heat in air}}$ mercury + sulphur dioxide
Reduction with carbon	extracting lead from lead(II) oxide	lead(II) oxide + carbon $\xrightarrow{\text{heat}}$ lead + carbon dioxide
	extracting iron from haematite (containing iron(III) oxide) in a blast furnace	carbon monoxide reacts with iron(III) oxide in the ore, producing iron iron(III) oxide + carbon monoxide $\xrightarrow{\text{heat}}$ iron + carbon dioxide
Electrolysis	extracting aluminium from bauxite (containing aluminium oxide)	aluminium forms at the negative electrode and oxygen gas bubbles off from the positive electrodes

3 The sequence of discovery of various metals relates closely with the ease of extracting the metals from their ores. Metals that are difficult to extract were discovered later.

4 The following table summarizes the years of discovery and extraction methods of some common metals.

Metal	Year of discovery	Ore	Main metallic compound in the ore	Extraction method
Gold (Au)	about 8000 B.C.	as free element	free element of gold	mechanical separation
Silver (Ag)		argentite or as free element	silver sulphide (Ag_2S) or free element of silver	displacement from solution◀ or mechanical separation
Mercury (Hg)		cinnabar	mercury(II) sulphide (HgS)	sulphide $\xrightarrow{\text{heat in air}}$ metal

◀ We will discuss displacement reactions in Unit 11.

Continued on next page

Metal	Year of discovery	Ore	Main metallic compound in the ore	Extraction method
Copper (Cu)	about 6000 B.C.	copper pyrite	copper(II) iron(II) sulphide ($CuFeS_2$)	sulphide $\xrightarrow[\text{heat in air}]{\text{controlled}}$ metal
Lead (Pb)	about 3500 B.C.	galena	lead(II) sulphide (PbS)	sulphide $\xrightarrow{\text{roast in air}}$ oxide, then oxide $\xrightarrow{\text{heat with carbon}}$ metal
Iron (Fe)	about 1200 B.C.	haematite	iron(III) oxide (Fe_2O_3)	oxide $\xrightarrow{\text{heat with carbon}}$ metal
Zinc (Zn)	1746	zinc blende	zinc sulphide (ZnS)	similar to that of lead
Aluminium (Al)	1826	bauxite	hydrated aluminium oxide ($Al_2O_3 \cdot 2H_2O$)	electrolysis of molten ore
Magnesium (Mg)	1808	magnesite	magnesium carbonate ($MgCO_3$)	
Sodium (Na)	1807	rock salt	sodium chloride (NaCl)	
Potassium (K)	1807	carnallite	hydrated potassium magnesium chloride ($KCl \cdot MgCl_2 \cdot 6H_2O$)	

5 Recycling is very important because

a) metal resources are limited;

b) it can save fuel and other resources;

c) it can reduce environmental impact; and

d) it can raise public awareness of conservation.

6 Metal resources are limited. The conservation methods are Reuse, Reduce, Recycle and Replace.

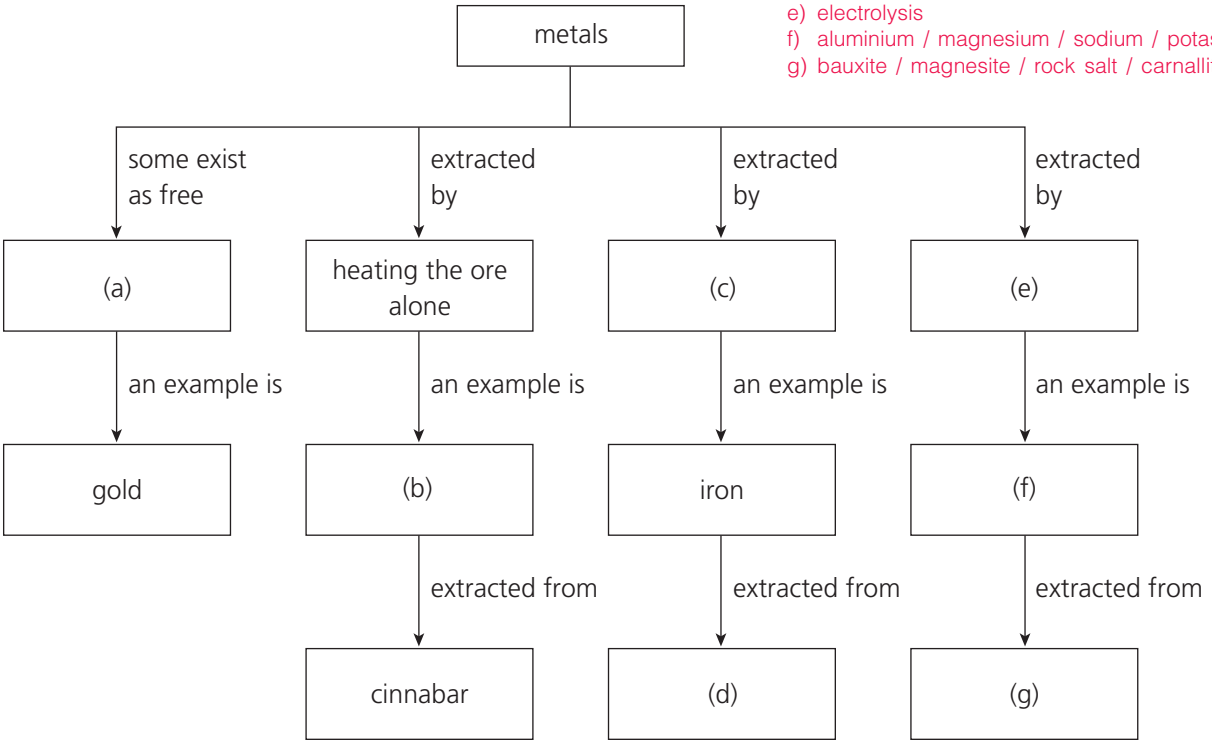
Unit Exercise

Note: The symbol ! indicates the level of difficulty of a question.

Part I Knowledge and understanding

1 Complete the following concept map.

a) elements
b) mercury
c) reduction with carbon
d) haematite
e) electrolysis
f) aluminium / magnesium / sodium / potassium
g) bauxite / magnesite / rock salt / carnallite



2 Complete the following description of the extraction of lead from lead(II) sulphide.

a) Lead exists as lead(II) sulphide in its ore called galena .

b) During the extraction process, the lead(II) sulphide is first changed to lead(II) oxide by roasting. Then carbon is used for the reduction.

c) The processes can be represented by the word equations:

lead(II) sulphide + oxygen $\xrightarrow{\text{roast in air}}$ lead(II) oxide + sulphur dioxide

lead(II) oxide + carbon $\xrightarrow{\text{heat}}$ lead + carbon dioxide

3 Complete the following description of the extraction of iron in a blast furnace.

a) During the extraction, a mixture of iron ore , coke and limestone is added from the top of the furnace.

b) At the same time, hot air is blown in at the bottom of the furnace. A chain of chemical reactions occurs. Carbon monoxide is produced in the process.

c) The carbon monoxide reacts with iron(III) oxide in the iron ore, producing iron.

The process can be represented by the word equation:

iron(III) oxide + carbon monoxide $\xrightarrow{\text{heat}}$ iron + carbon dioxide

4 Complete the following description of the extraction of aluminium.

a) The most common ore of aluminium is called bauxite . It consists mainly of aluminium oxide .

b) The method used to extract aluminium from its ore is electrolysis . The electrodes are made of graphite .

c) During the process, aluminium forms at the negative electrode while oxygen gas bubbles off from the positive electrodes.

5 The pictures show some uses of metals.

Aircraft bodies



Electric wiring



Railway tracks



Complete the table.

Use	Name of metal with this use	Property on which the use depends
Aircraft bodies	aluminium	low density
Electric wiring	copper	good conductor of electricity
Railway tracks	iron	strong

(Edexcel IGCSE (Higher Tier), Chemistry, Paper 2, Jun. 2009, 1)

Part II Multiple choice questions

6 Which of the following combinations concerning the uses of metals is correct?

	Metal	Use
(A)	copper	making cooking utensils
B	iron	coating for other metals
C	titanium	making electric wires
D	zinc	making cans for canned food

Option D — Cans for canned food are made of mild steel coated with tin.

7 Which of the following statements concerning the extraction of aluminium from its oxide by electrolysis is correct?

Refer to page T28 for explanations.

A The process is carried out in an aluminium tank.
(B) The graphite lining of the tank acts as the negative electrode.
C Aluminium forms at the positive electrode.
D Oxygen forms at the negative electrodes.

8 Which of the following metals was NOT discovered in ancient times?

A Copper
B Gold
(C) Sodium — Sodium was discovered in 1807.
D Tin

9 In which of the following ores are the main metallic compounds sulphides?

(1) Argentite
(2) Copper pyrite
(3) Zinc blende

A (1) and (2) only
B (1) and (3) only
C (2) and (3) only
(D) (1), (2) and (3)

Ore	Main metallic compound in the ore
Argentite	silver sulphide
Copper pyrite	copper(II) iron(II) sulphide
Zinc blende	zinc sulphide

Metal	Extraction method
Copper	controlled heating in air
Magnesium	electrolysis of molten ore
Silver	displacement from solution or mechanical separation

10 Which of the following combinations are correct?

	Metal	Extraction method
(1)	copper	controlled heating in air
(2)	magnesium	electrolysis of molten ore
(3)	silver	heating with carbon

(A) (1) and (2) only
B (1) and (3) only
C (2) and (3) only
D (1), (2) and (3)

11 Which of the following about iron is / are correct?

(1) It is the most abundant metal in the Earth's crust.
(2) It is obtained from its ore by carbon reduction.
(3) It resists corrosion.

A (1) only
(B) (2) only
C (1) and (3) only
D (2) and (3) only

(1) Aluminium is the most abundant metal in the Earth's crust.
(3) Iron has very low corrosion resistance.

12 Which of the following statements about copper are correct?

(1) It can be obtained from its sulphide by heating in air.
(2) It is a very good conductor of electricity.
(3) It is extensively recycled.

(A) (1) and (2) only
B (1) and (3) only
C (2) and (3) only
D (1), (2) and (3)

13 In which of the following experiments would a metal be produced?

Refer to page T28 for explanations.

(1) Heating haematite
(2) Heating mercury(II) sulphide
(3) Mixing lead(II) oxide with carbon powder

A (1) only
(B) (2) only
C (1) and (3) only
D (2) and (3) only

Part III Structured questions

14 The table below summarizes the extraction process for iron.

	Iron
Common ore	haematite
Energy source used in the extraction	combustion of coke in air
Substances used in the extraction	limestone, coke

During the extraction of iron, a chain of chemical reactions occurs.

Stage I The coke reacts with oxygen in the air to give carbon dioxide.

Stage II The limestone is decomposed by heat.

Stage III The carbon dioxide produced reacts with more coke, giving carbon monoxide.

Stage IV The carbon monoxide reacts with the main metallic compound in the iron ore to give iron.

a) Name the main metallic compound in haematite.
Iron(III) oxide

b) Write a word equation for the reaction that occurs in *Stage II*.
calcium carbonate $\xrightarrow{\text{heat}}$ calcium oxide + carbon dioxide

c) Write a word equation for the reaction that occurs in *Stage IV*.
iron(III) oxide + carbon monoxide $\xrightarrow{\text{heat}}$ iron + carbon dioxide

15 The table below shows some of the properties of five metals.

Metal	Density ($g\ cm^{-3}$)	Melting point (°C)	Relative hardness	Relative electrical conductivity	Relative heat conductivity
Chromium	7.2	1 857	8.5	0.8	0.9
Cobalt	8.9	1 495	5.0	1.7	1.0
Copper	9.0	1 085	3.0	6.0	4.1
Nickel	8.9	1 453	4.0	1.4	0.9
Zinc	7.1	420	2.5	1.7	1.2

a) The diagram below shows an electric wire.



Use information from the table to explain why copper is the most suitable metal to make electric wires. Copper has the highest electrical conductivity.

b) This drill bit is used to make holes in metals.



Which metal would be the most suitable to make a drill bit?

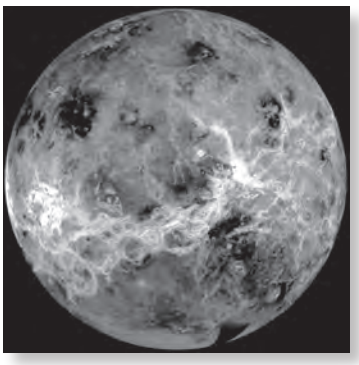
Choose from the table. Explain your answer.
Chromium. It is the hardest metal / has the highest melting point.

c) Metals are good electrical conductors because a charged particle can move.

What is the name of this charged particle? Electron

(OCR GCSE Gateway Science (Higher Tier), Chem. B, Unit 1, Jan. 2009, 13)

16 The photograph shows the planet Venus.



Although Venus is similar in size to the Earth, it is very different in other ways.

The temperature at the surface of Venus is about 470 °C. The atmospheric pressure is 90 times that of the Earth.

The clouds in the atmosphere of Venus are made up of droplets of sulphuric acid.

The table lists some properties of metals that could be used to make a space probe to land on Venus.

Metal	Melting point (°C)	Relative density	Reaction with sulphuric acid
Copper	1 083	8.9	no reaction
Lead	328	11.3	no reaction
Magnesium	650	1.7	fizzes vigorously
Nickel	1 453	8.9	fizzes slowly
Titanium	1 675	4.5	no reaction
Zinc	420	7.1	fizzes quite vigorously

The probe needs to be launched with enough energy to escape the Earth's gravity. To make this easier, the mass of the probe needs to be as low as possible. The probe also needs to withstand the conditions on the surface of Venus.

Use the information in the table to answer parts (a) to (c).

a) i) Which metal in the table could be used to make a probe with the lowest density?
Magnesium

ii) Why would this metal be UNSUITABLE for making a probe to land on Venus?
It would react with sulphuric acid.

b) Very small amounts of lead can be used in electric circuits.

Why would lead NOT be suitable for use in the electric circuits of a probe to land on Venus?
The temperature on Venus is higher than the melting point of lead.

c) Choose a metal from the table that would be the most suitable for making a probe to land on Venus. Give TWO reasons for your choice.

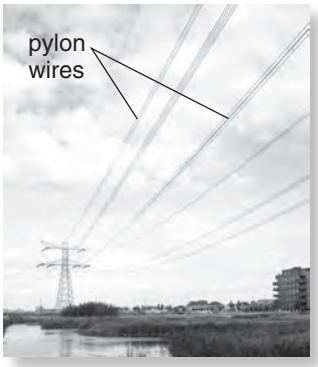
(Edexcel IGCSE, Chemistry, Paper 2C, Jun. 2011, 3)

16 c) Titanium
Any two of the following:
- It has a low density.
- It has a high melting point.
- It does not react with sulphuric acid.

17 The table below shows the properties of some metals.

Metal	Melting point (°C)	Density (g cm^{-3})	Relative electrical conductivity	Cost per tonne (£)
Aluminium	660	2.7	40	1 350
Copper	1 083	8.9	64	3 800
Iron	1 535	7.9	11	400
Silver	962	10.5	67	20 000
Zinc	420	7.1	18	870

a) Aluminium is used to make pylon wires.



Their densities are too high (so the wires would sag).
Silver and copper are too expensive.

Silver and copper are better electrical conductors than aluminium. However, silver and copper are NOT used to make pylon wires. Explain why silver and copper are NOT used to make pylon wires. Use the table to help you.

b) Which metal from the table would be the best to use for a door stop for keeping doors open? Write down TWO reasons to explain why.



c) Metals are good conductors of electricity. Use ideas about the structure of metals to explain how metals conduct electricity. The delocalized electrons can move.

(OCR GCSE Gateway Science (Higher Tier), Additional Science B, Unit 1, B623/02, Jan. 2010, 6)

17 b) Copper
Any two of the following:
- It has a high density.
- It is lustrous / shiny.
- It is relatively cheap.
- It does not rust.

Unit Exercise

18 The flow diagram shows the main stages used to extract a metal from its ore.

mining the ore → purifying the ore → extracting the metal

The table shows information about three metals.

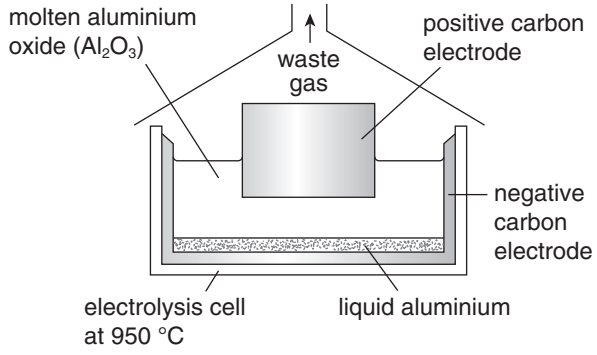
Metal	Metal ore	Purified ore	% of metal in the ore	% of metal in the Earth's crust
Aluminium	bauxite	aluminium oxide, Al_2O_3	28.0	8.0
Copper	chalcocite	copper(I) sulphide, Cu_2S	0.5	0.001
Iron	haematite	iron(III) oxide, Fe_2O_3	29.0	5.0

a) Use the information in the table and your knowledge and understanding to help you to answer the questions.

i) Suggest why purifying the copper ore produces large quantities of waste.
Very low percentage of the metal in the ore.

ii) Suggest why the annual world production of iron is forty times greater than that of aluminium.

b) Aluminium is used for drinks cans. Aluminium is extracted from its purified ore by electrolysis.



18 a) ii) Any one of the following:
- Iron uses less energy for extraction.
- Iron has more uses.
- There is more demand for iron.
- Iron is stronger.
- Iron is cheaper.
- Iron is easier to extract.

i) Suggest why the aluminium produced in the electrolysis cell is a liquid.
The melting point of aluminium is lower than 950 °C.

ii) In this electrolysis, aluminium and oxygen gas are produced from the aluminium oxide.

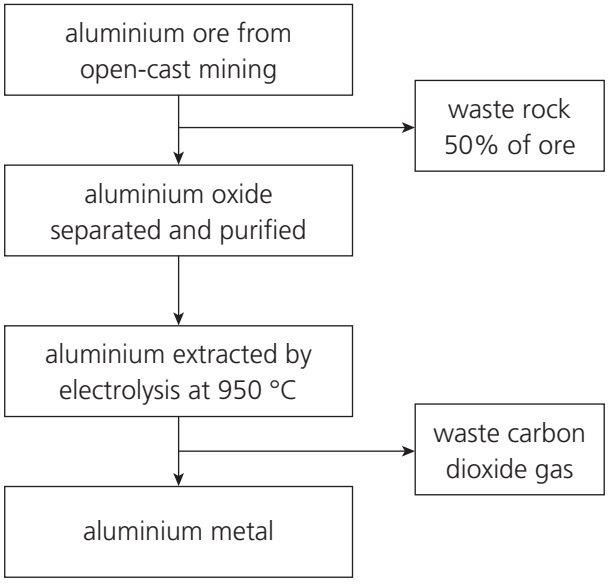
Use the information in the diagram to suggest why most of the waste gas is carbon dioxide and NOT oxygen.

(AQA GCSE (Higher Tier), Chemistry, Jan. 2011, Unit 1, 1(a)–(b))

18 b) ii) The electrodes are made of carbon. Oxygen reacts with the electrodes to give carbon dioxide.

19 Aluminium has many uses because of its low density, good electrical conductivity, flexibility and resistance to corrosion.

The main steps in the extraction of aluminium are shown in the flow chart.



Use the information in the flow chart to suggest the benefits of recycling aluminium.

(AQA GCSE (Higher Tier), Chemistry, Unit 1, Jan. 2008, 4(a))

19 Any three of the following:
- Resources are conserved.
- Less energy is used in recycling aluminium than in extracting the metals from ores.
- Less mining / associated environmental problems (dust / loss of habitat)
- Less waste rock / landfill
- Less carbon dioxide